

Assessment 1: Functions and Polynomial Review

75-minute practice based on Assessment 1 functions and polynomial review

Name: _____

Date: _____

Time: 75 minutes | Total: 100 points

Before you begin

- Show your steps. Answers without supporting work may earn partial credit.
- Use interval notation for domains and intervals of increase or decrease.
- For graphs, label important points and show whether the graph crosses or touches.
- Suggested section times add to 75 minutes. Keep moving and return to a hard item later.

Section A: Function Basics

18 minutes | 23 points

1 Read a function from a table

5 points

The table lists four ordered pairs:

x	-2	0	0	3
y	5	-1	-1	4

Does the relation define a function? Use the repeated x-value to explain your answer, then give the range.

2 Catch the missing domain restriction

8 points

Let $f(x) = \sqrt{5 - x}$ and $g(x) = x^2 - 1$. A student says the domain of $(f/g)(x)$ is all $x \leq 5$.

Write the quotient, identify every excluded input the student missed, and give the domain in interval notation.

3 Find and compare two compositions**10 points**

Let $A(x) = 2x - 1$ and $B(x) = x^2 + 3$.

(a) Find and simplify $B(A(x))$.

(b) Evaluate $B(A(4))$.

(c) Evaluate $A(B(4))$ and compare it with your answer to part (b).

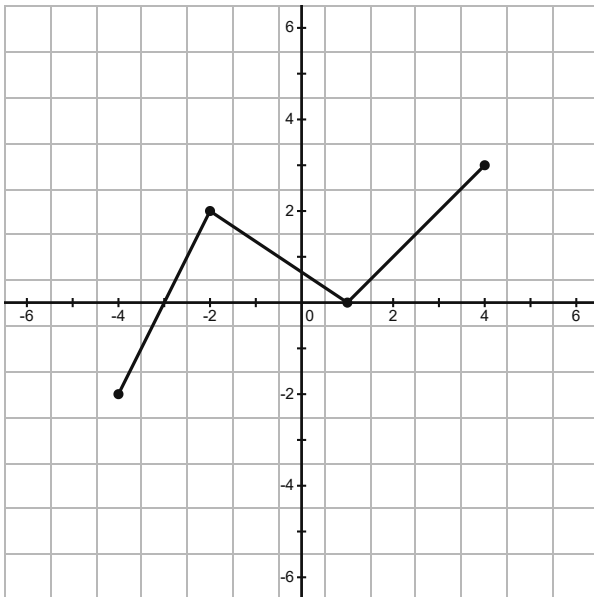
Section B: Transformations and Piecewise Functions

22 minutes | 28 points

4 Graph both orders of a composition

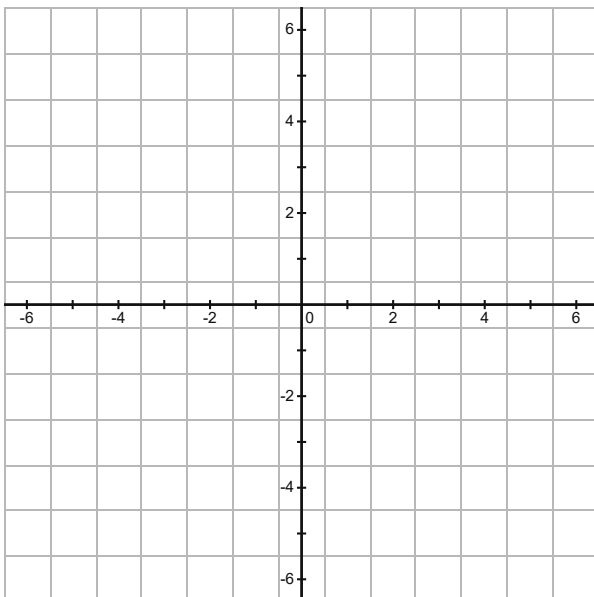
10 points

The graph of f is the connected line graph shown. Let $g(x) = 2x - 2$.



On the grid below, graph both $f(g(x))$ and $g(f(x))$. Use a solid line for the first graph and a dashed line for the second. Mark and label every transformed vertex, including endpoints.

Each square is 1 unit. The axes are numbered every 2 units.



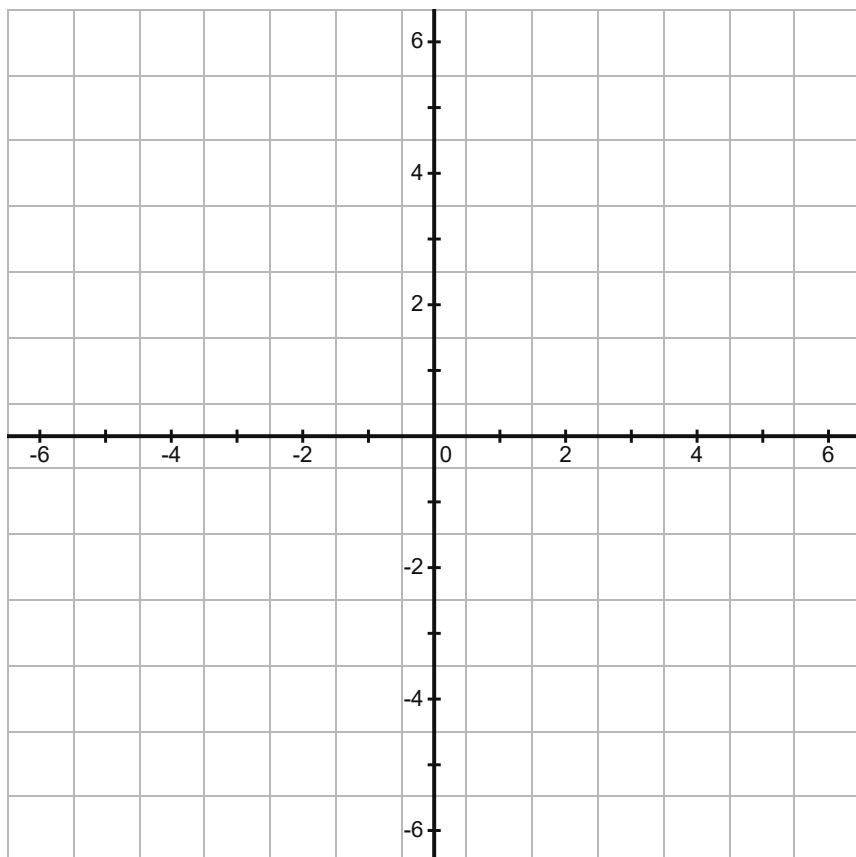
5**Evaluate and graph a piecewise rule****8 points**

$$\text{Let } p(x) = \begin{cases} 2 - x & x < 1 \\ x + 2 & x \geq 1 \end{cases}.$$

(a) Find $p(0)$, $p(1)$, and $p(3)$.

(b) Graph both pieces. Show the open and closed endpoint at the boundary $x = 1$.

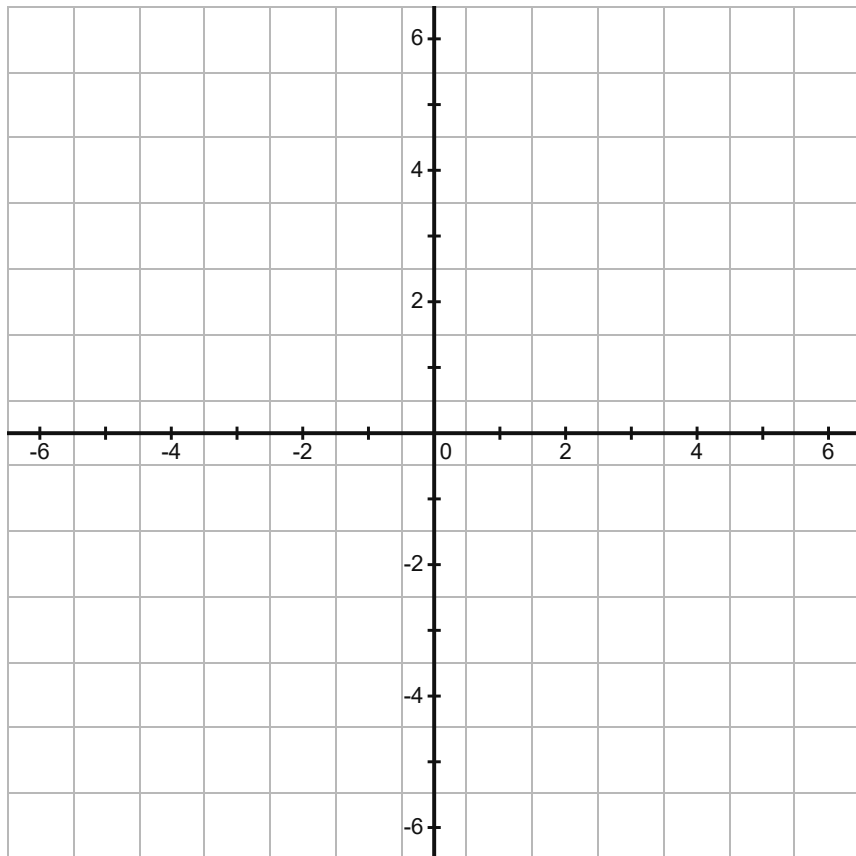
Each square is 1 unit. The axes are numbered every 2 units.



6**Graph a square-root translation****10 points**

Translate the parent graph $y = \sqrt{x}$ left 4 units and down 3 units. Write its rule, then graph it. Mark its starting point and three more points. State its domain and range, and check whether $(-3, -2)$ lies on the graph.

Each square is 1 unit. The axes are numbered every 2 units.



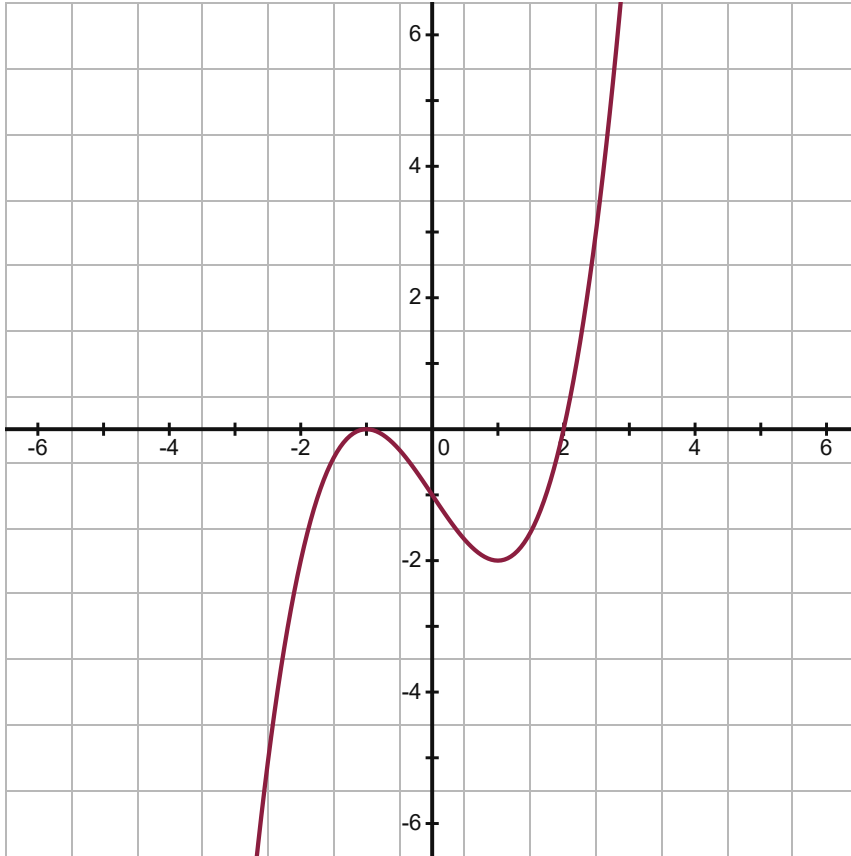
Section C: Polynomial Structure and Graphs

20 minutes | 27 points

7 Recover a polynomial from its graph

7 points

The graph is a cubic polynomial with a repeated zero. Use its intercepts and turning points to write an equation in factored form. Then state where it is increasing and decreasing, and give its relative maximum and minimum points.



8**Choose the polynomial that fits the graph behavior****10 points**

A polynomial graph crosses the x -axis at -2 , touches and turns at 1 , falls to the left, and rises to the right. Which rule could represent it?

(a) Choose one and explain how its zeros, multiplicities, and leading term match.

A. $(x + 2)(x - 1)^2$

B. $-(x + 2)^2(x - 1)$

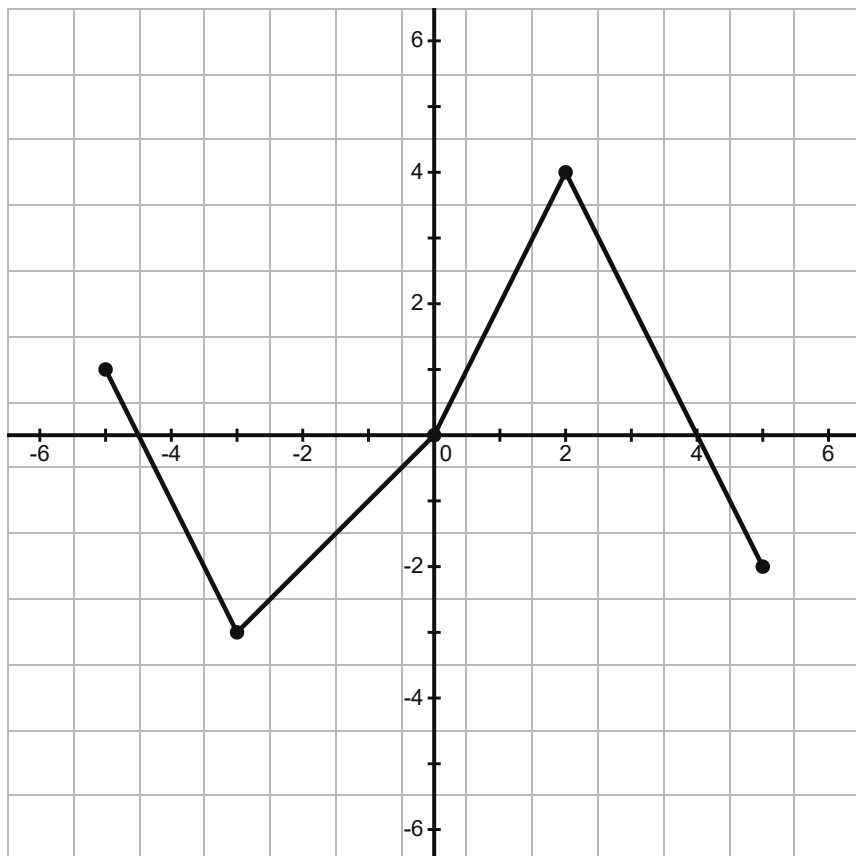
C. $(x + 2)^2(x - 1)^2$

D. $-(x + 2)(x - 1)^2$

9 Check a classmate's graph analysis**10 points**

The graph of r is made of connected line segments through the plotted vertices. A classmate reports: domain $[-5, 5]$, range $[-3, 4]$, zeros -4 , 0 , and 4 ; increasing on $(-5, -3)$ and $(2, 5)$; relative maximum $y = 4$ and relative minimum $y = -3$.

Check each part of the report and correct any errors. Give the increasing and decreasing intervals, and write the relative maximum and minimum as coordinate points.



Section D: Function Models and Transformations

15 minutes | 22 points

10 Choose a valid composition using units

12 points

A rental shop charges $D(d) = 9d + 12$ dollars for d days. A tax function converts a pre-tax price c dollars to $T(c) = 1.08c$ dollars.

(a) Write the composition that gives the after-tax rental cost, and explain why its order fits the units.

(b) Simplify the composite rule.

(c) Find the total cost for a 4-day rental.

(d) Explain why reversing the order does not model this situation.

11 Keep the original domain when composing

10 points

Let $f(x) = \sqrt{x + 1}$ and $g(x) = x^2 - 1$.

(a) Find and simplify $g(f(x))$. State its domain, and explain why simplifying the square root does not make the original domain larger.

(b) Find and simplify $f(g(x))$. State its domain. Explain why its domain differs from part (a).
